

碩士學位論文

Transformer-Diffusion-PPO를
이용한 다중 에이전트 강화 학습 연구

A Study on Multi-Agent
Reinforcement Learning using
Transformer-Diffusion-PPO

國立韓國交通大學校 大學院

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Abstract

Title: A Study on Multi-Agent Reinforcement Learning using Transformer-Diffusion-PPO

Multi-agent reinforcement learning (MARL) is a key technique for cooperative autonomous systems, but it remains difficult when agents must act from incomplete local observations while coordinating in a shared continuous-control task. In such settings, each agent observes only part of the environment, the apparent dynamics are non-stationary because other agents are learning at the same time, and simple unimodal action policies may be insufficient for coordinated behavior. This thesis studies these challenges in a field-of-view (FOV)-limited multi-agent control setting and proposes a Multi-Agent Transformer-Diffusion-PPO (MATDP) reinforcement learning framework.

The proposed framework formulates the task as a decentralized partially observable Markov decision process. Each actor receives a short history of local observations masked by a FOV radius, while a centralized critic uses unmasked observations during training. The actor first encodes each agent's observation history with a Transformer encoder and then applies a team-level Transformer to produce a cross-agent latent representation. Conditioned on this representation, a joint diffusion policy generates the 15-dimensional action vector of all three agents together. The policy is optimized online with Proximal Policy Optimization

(PPO), using step-average diffusion log probabilities, a centralized value function, potential-based reward shaping, and an assignment auxiliary objective for training stabilization.

Experiments are conducted on the MPE `simple_spread` benchmark with three agents and three landmarks. The main evaluation uses a four-stage flow that gradually reduces the actor FOV radius from 2.0 to 1.5, 1.0, and 0.5. The proposed MATDP model is compared with MATD, which removes online PPO updates, and MATG, which replaces the diffusion policy with a Gaussian policy, under the same training data and flow setting. In the main experiments, MATDP achieves the highest mean success rate over the final 100 episodes at every FOV stage, obtaining 0.190, 0.170, 0.150, and 0.150, respectively. MATD performs substantially worse in the hardest stage, showing that online policy adaptation is necessary.

The results also identify a clear limitation. In the extreme FOV=0.5 condition, MATDP can still produce stochastic successful episodes, but deterministic evaluation remains weak. Additional diagnostics show that actor-side memory and stronger exploration provide only limited improvement, suggesting that the main bottleneck is insufficient information flow under severe partial observability. Therefore, this thesis concludes that Transformer-Diffusion policies are promising for partially observable multi-agent control, while robust deterministic coordination in extreme visibility-limited settings requires future work on communication, stronger memory, and richer evaluation environments.

Keywords: multi-agent reinforcement learning, transformer, diffusion policy, proximal policy optimization, partial observability, centralized training decentralized execution

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I. Introduction

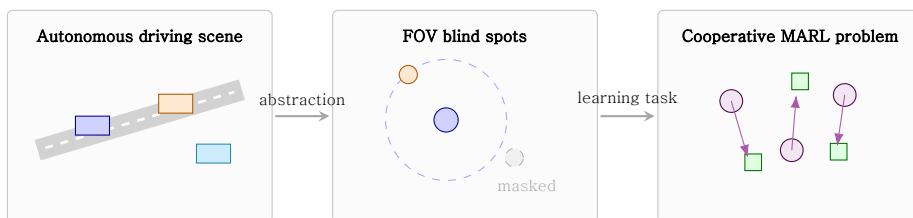
1. Background and Motivation

Reinforcement learning (RL) provides a general framework for learning sequential decision-making policies through interaction with an environment [1]. Deep reinforcement learning (DRL) extends this framework by using neural networks to represent policies and value functions, making it possible to solve high-dimensional control tasks that are difficult to model manually. Early successes such as deep Q-networks on Atari games demonstrated that neural policies can learn complex behavior from experience [2, 3]. For continuous control, policy-gradient and actor-critic methods such as TRPO and PPO further improved the stability of neural policy optimization [4, 5].

Many real control problems, however, are not single-agent problems. Autonomous vehicles, mobile robots, and distributed sensing systems often contain multiple decision makers that act in the same physical or virtual environment. In such systems, the action of one agent changes the observations and rewards of the others, so each agent faces a non-stationary learning problem. Multi-agent reinforcement learning (MARL) studies this setting and has become an important direction for cooperative and competitive control [6, 7]. The difficulty becomes greater when agents cannot observe the full state. Under partial observability, a local observation may not contain enough information to identify the true global configuration, and the policy must infer hidden information from observation history and coordination patterns.

This thesis focuses on cooperative multi-agent continuous control under partial observability. The experimental task is based on the MPE `simple_spread` environment, where three agents must cover three landmarks while avoiding inefficient overlap. The task is compact, but it contains several important challenges: agents need to coordinate, actions are continuous-valued through the environment interface, and observation quality can be controlled by applying a field-of-view (FOV) mask. By reducing the FOV radius, the task becomes a testbed for studying how a policy adapts when local perception becomes progressively weaker.

Figure 1.1 summarizes this motivation by linking a multi-agent driving-like scenario to the FOV-limited cooperative control problem studied in this thesis. The left panel represents a shared-space driving scene in which multiple vehicles must coordinate. The middle panel abstracts the effect of FOV-induced blind spots, where relevant objects may be hidden from the local observation. The right panel converts this situation into a cooperative MARL problem in which agents must learn coordinated actions from partial histories.



<Figure 1.1> Motivation of the thesis

2. Problem Statement

The problem considered in this thesis can be described as a decentralized partially observable Markov decision process

(Dec-POMDP) [8]. At each time step, each agent receives only a local observation instead of the complete environment state. The agents then choose a joint action, the environment transitions to a new state, and a team reward is produced. The goal is to learn decentralized actor behavior that can coordinate all agents, even though each actor operates from limited local information.

Three technical challenges motivate the proposed approach. First, a memoryless policy may fail when important landmarks or other agents are temporarily outside the FOV. A useful actor should therefore model recent observation history rather than relying only on the current frame. Second, multi-agent coordination requires reasoning about joint behavior. If each agent samples its action independently, the resulting joint action may contain conflicts or redundant movements. Third, a conventional Gaussian policy may be too restrictive for multi-modal action distributions, especially when several coordinated movements are plausible from the same partial observation.

To address these challenges, this thesis proposes a Multi-Agent Transformer-Diffusion-PPO (MATDP) framework. The actor uses Transformer-based sequence encoding to summarize each agent's recent observation history, a team-level Transformer to build a cross-agent latent representation, and a joint diffusion policy to generate the 15-dimensional action vector of all agents. PPO is used for online policy adaptation, while the critic follows the centralized-training decentralized-execution (CTDE) principle by using unmasked observations during training.

3. Research Objectives

The main objective of this thesis is to evaluate whether a Transformer-Diffusion policy can improve cooperative multi-agent control under progressively stronger partial observability. More specifically, the study aims to answer the following questions:

- Can observation-history modeling with Transformers help agents act under FOV-limited local observations?
- Does a joint diffusion action policy provide a better success-rate profile than a simpler Gaussian action policy under the same training protocol?
- Is online PPO adaptation necessary after assignment-based pretraining, or can a pretrained diffusion policy alone handle the FOV flow?
- What failure modes appear when the FOV becomes extremely small, and what do they imply for future multi-agent control systems?

The thesis deliberately evaluates the method in a bounded and reproducible way. The main evidence comes from controlled experiments with the same training budget, flow schedule, and environment configuration. Additional diagnostic experiments are used to analyze the hardest $\text{FOV}=0.5$ condition, but they are not treated as direct replacements for the main flow comparison.

4. Main Contributions

The contributions of this thesis are summarized as follows.

- A Dec-POMDP formulation of the FOV-limited MPE `simple_spread` task, where actor observations are masked but critic observations remain unmasked during training.
- A MATDP actor architecture that combines per-agent Transformer history encoding, team-level cross-agent encoding, and 15-dimensional joint diffusion action generation.
- A PPO training procedure for joint diffusion policies, using step-average diffusion log probabilities and centralized value estimation.
- A practical stabilization design based on potential-based reward shaping and assignment auxiliary learning derived from task geometry.
- A four-stage FOV flow evaluation showing that MATDP achieves the highest final-100-episode success rate among the compared methods, while also identifying that the extreme FOV=0.5 case remains unresolved in deterministic evaluation.

5. Thesis Organization

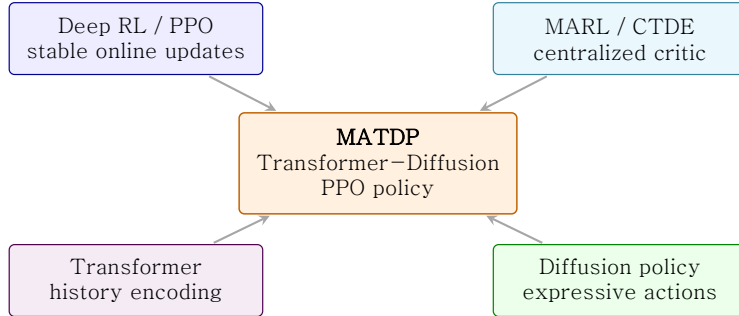
The rest of this thesis is organized as follows. Chapter 2 reviews the related work on deep reinforcement learning, multi-agent reinforcement learning, Transformer-based sequence modeling, diffusion policies, PPO, and reward shaping. Chapter 3 presents the proposed MATDP methodology, including the Dec-POMDP formulation, actor and critic architectures, diffusion policy optimization, and training stabilization mechanisms. Chapter 4 reports the experimental setup, main flow results,

ablation studies, and extreme partial-observability analysis. Chapter 5 concludes the thesis, summarizes the main findings, and discusses future research directions.

II. Related Work

The method developed in this thesis is shaped by a specific control problem rather than by a single research line. Agents must act continuously, receive only field-of-view-limited observations, coordinate with teammates, and learn from sparse multi-agent feedback. For this reason, the related work is organized around four design requirements: stable online policy optimization, centralized training for cooperative MARL, sequence modeling under partial observability, and expressive continuous action generation. Reward shaping is reviewed separately as a practical stabilization component for difficult exploration.

Figure 2.1 shows how these literature strands connect to the final MATDP design, so the chapter proceeds from optimization and MARL foundations to sequence modeling, diffusion policies, and stabilization. Each outer block corresponds to one requirement of the target problem, and the central block indicates how these requirements are integrated into the proposed Transformer-Diffusion-PPO framework.



<Figure 2.1> Design requirements of MATDP

1. Deep Reinforcement Learning and Policy Optimization

Reinforcement learning studies how an agent improves a policy through interaction with an environment. At each step, the agent observes a state or local observation, chooses an action, receives a reward, and updates the policy to increase expected cumulative return. Deep reinforcement learning replaces tabular representations with neural networks, which makes it possible to learn nonlinear policies from high-dimensional or structured observations.

Early deep RL success was largely demonstrated in discrete action domains. Deep Q-Networks (DQN) showed that neural value functions can learn strong policies in Atari games [2, 3]. The task studied in this thesis is different because the MPE interface is treated as continuous multi-agent control: each agent produces movement-related action channels, and the proposed actor generates a 15-dimensional joint action for three agents. Actor-critic methods are therefore more relevant than value-only methods. DDPG introduced deterministic actor-critic learning for continuous control [9], while TRPO used a trust-region constraint to avoid overly large policy updates [4].

PPO provides a practical compromise between update stability and implementation simplicity [5]. Instead of solving a full trust-region problem, PPO clips the probability ratio between the new and old policy. Advantage estimates are commonly computed with Generalized Advantage Estimation (GAE), which controls the bias-variance trade-off through an exponentially weighted estimator [10]. Given an advantage estimate $\hat{A}_t$,

the clipped objective is

$$L^{CLIP}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right], \quad (2.1)$$

where $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$. This objective is used as the optimization basis of MATDP, but the policy distribution is not a standard Gaussian. Because the actor is a diffusion policy, Chapter III extends the PPO ratio to a step-average form over the denoising chain.

2. Multi-Agent Reinforcement Learning under Partial Observability

Multi-agent reinforcement learning extends RL to settings where several agents learn and act in a shared environment. Even in cooperative tasks, this creates two difficulties that do not appear in the same form in single-agent learning. First, each agent faces a changing environment because other agents are also updating their policies. Second, a team reward must be assigned back to individual or joint behaviors. These issues are important in cooperative navigation, mobile robot teams, traffic systems, and connected autonomous vehicles [7].

The Dec-POMDP framework formalizes cooperative decision-making under uncertainty [8]. In a Dec-POMDP, each agent receives only partial observations and selects actions from its observation history. This directly matches the field-of-view setting in this thesis: when a landmark or teammate is outside the FOV radius, it is removed from the actor input, although it may still matter for the global task outcome.

Centralized training with decentralized execution (CTDE) is a common

response to this mismatch between local perception and global reward. During training, a centralized critic can use global or less restricted information to estimate values, while decentralized actors are still trained to act from local observations at execution time. MADDPG is an influential CTDE actor-critic method for mixed cooperative-competitive environments [6]. COMA and QMIX address related credit-assignment and value-decomposition problems [11, 12]. More recent studies show that PPO-style methods can also be strong cooperative MARL baselines when implementation details are carefully controlled [13].

MATDP follows the CTDE principle but keeps the actor information bottleneck explicit. The critic receives unmasked observations during training, whereas each actor receives FOV-masked histories. This distinction is important: the critic can make learning less noisy, but it cannot give hidden objects to the actor at test time. The remaining gap between critic knowledge and actor perception is one of the main reasons why the extreme FOV=0.5 stage remains difficult in Chapter IV.

Another line of MARL work introduces explicit communication between agents [14]. Communication is not used in the present method. This choice makes the experiment cleaner for testing history encoding and joint action generation, but it also limits what the actors can infer when the visible fraction becomes very small.

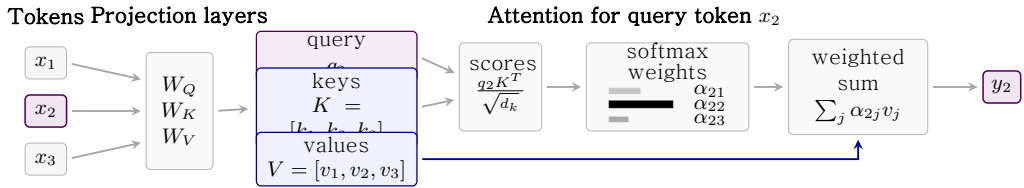
3. Transformer-Based Sequence Modeling

Transformers were introduced for sequence modeling through self-attention [15]. Instead of processing a sequence only through recurrent state updates, self-attention computes relationships between

tokens and allows the model to select relevant parts of the input. For query, key, and value matrices Q , K , and V , scaled dot-product attention is defined as

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V. \quad (2.2)$$

Figure 2.2 illustrates this operation for one query token. The visual emphasis is on the fact that one query can assign different weights to all key-value pairs, which is the mechanism later used to summarize FOV-masked histories.



<Figure 2.2> Scaled dot-product attention for query token x_2

In reinforcement learning, sequence modeling is useful because the current observation may not contain enough information for action selection. Decision Transformer showed that offline RL trajectories can be modeled as token sequences conditioned on return information [16]. Multi-agent sequence models further indicate that temporal history and agent identity can both be useful when observations are structured across agents [17].

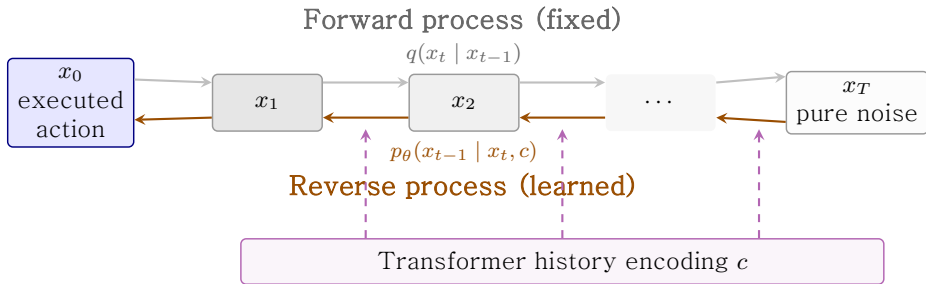
The Transformer use in MATDP is narrower and more online than in return-conditioned offline RL. A per-agent Transformer summarizes a fixed-length observation history for each actor, and a team-level Transformer then processes the resulting agent embeddings. The goal is not to predict an entire trajectory, but to provide a compact latent representation for PPO-controlled action generation under partial

observability. The FOV mask is applied before encoding, so the Transformer cannot recover invisible objects directly; it can only exploit temporal evidence that has appeared in the actor's local history.

4. Diffusion Policies for Continuous Action Generation

Diffusion models learn to reverse a gradual noising process [18]. When used as policies, they start from noise and iteratively denoise it into an action conditioned on an observation or latent state. This gives the policy a richer action-generation mechanism than a unimodal Gaussian distribution, which can be useful when several coordinated action patterns are possible.

Figure 2.3 summarizes the forward noising process and the learned reverse denoising process in the action-generation setting used by this thesis. The upper row represents the fixed forward process $q(x_t|x_{t-1})$, while the lower row represents the learned reverse model $p_\theta(x_{t-1}|x_t, c)$. The conditioning variable c denotes the Transformer history encoding used to guide denoising.



<Figure 2.3> DDPM forward and reverse processes for action generation

Diffusion Policy applied this idea to visuomotor control and showed that denoising-based action generation can be effective for robotic

manipulation [19]. DPPO further connected diffusion policies with online policy optimization by adapting diffusion action models through reinforcement learning [20]. These studies motivate the use of diffusion for continuous control, but the multi-agent setting adds an additional design choice: whether actions should be sampled independently for each agent or generated jointly.

MATDP chooses joint generation. The diffusion policy outputs the concatenated 15-dimensional action of all three agents instead of three independent single-agent actions. This lets the action generator represent cross-agent dependencies inside one denoising process. The cost is that PPO cannot use a single Gaussian log probability; the implementation must store and recompute step-wise denoising log probabilities. This is why Chapter III defines a step-average diffusion ratio for policy updates.

5. Reward Shaping and Training Stabilization

The MPE `simple_spread` task provides dense distance-related rewards, but successful coordination under limited FOV can still be rare and unstable during early learning. This is especially true when the actor often observes no landmark or teammate. Potential-based reward shaping is useful here because it can provide additional learning signal while preserving the optimal policy under suitable conditions [21]. A shaping term can be written as

$$F(s, a, s') = \gamma\Phi(s') - \Phi(s), \quad (2.3)$$

where $\Phi(s)$ is a potential function and γ is the discount factor.

The shaping potential used in this thesis is based on soft assignment between agents and landmarks. Soft assignment is related to entropy-regularized optimal transport, where Sinkhorn-style normalization gives a differentiable alternative to hard matching [22]. This design encourages agents to move toward coverage configurations without replacing the environment reward. In addition, an assignment auxiliary objective supplies a supervised movement target derived from a minimum-cost agent-landmark assignment, which is commonly solved by the Hungarian method [23]. The auxiliary target is used for stabilization and pretraining; the final behavior still comes from the learned actor updated by PPO.

6. Chapter Summary

The reviewed work defines both the motivation and the boundaries of MATDP. PPO provides the online optimization backbone, CTDE provides a way to train from richer information than the actor can observe, Transformers encode short FOV-masked histories, diffusion policies generate coordinated continuous joint actions, and potential-based shaping improves exploration. At the same time, none of these components alone removes the actor's information bottleneck. The next chapter therefore describes how the components are combined and where the implementation departs from their standard forms.

III. Proposed Methodology

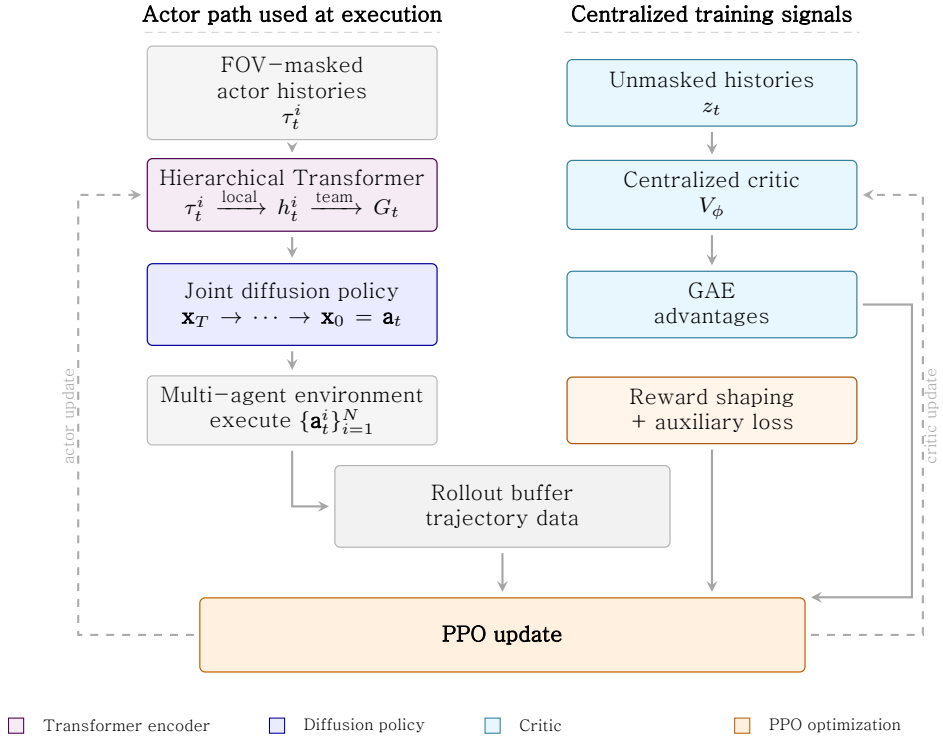
MATDP is designed for cooperative multi-agent continuous control when each actor has only a limited local view of the environment. In the implemented setting, the actor receives a field-of-view (FOV)-masked observation history, while the critic is trained with unmasked observations. The method combines per-agent Transformer history encoding, team-level cross-agent encoding, and a joint diffusion policy that generates the coordinated action of all agents.

Figure 3.1 gives the method-level overview of this design. It separates the execution-time actor path from the centralized training signals, which is the main organizational principle used throughout the chapter. At execution time, the actor uses only FOV-masked histories, Transformer encoding, and the joint diffusion policy to produce joint actions. During training, rollout data are stored in a central buffer, the critic and GAE provide value and advantage estimates, and reward shaping and assignment auxiliary losses enter the PPO update.

1. Problem Formulation

The cooperative control problem is formulated as a Decentralized Partially Observable Markov Decision Process (Dec-POMDP):

$$\mathcal{M} = \langle \mathcal{I}, \mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{R}, \Omega, \mathcal{O}, \gamma \rangle, \quad (3.1)$$



<Figure 3.1> Method-level overview of MATDP

where $\mathcal{I} = \{1, \dots, N\}$ is the set of agents, $\mathcal{S}$ is the global state space, $\mathcal{A} = \times_i \mathcal{A}_i$ is the joint action space, $\mathcal{T}$ is the transition function, $\mathcal{R}$ is the team reward, Ω is the joint observation space, $\mathcal{O}$ is the observation function, and γ is the discount factor.

At time step t , agent i receives a local observation o_t^i rather than the full state s_t . A fixed-length observation history is maintained:

$$\tau_t^i = (o_{t-H+1}^i, o_{t-H+2}^i, \dots, o_t^i), \quad (3.2)$$

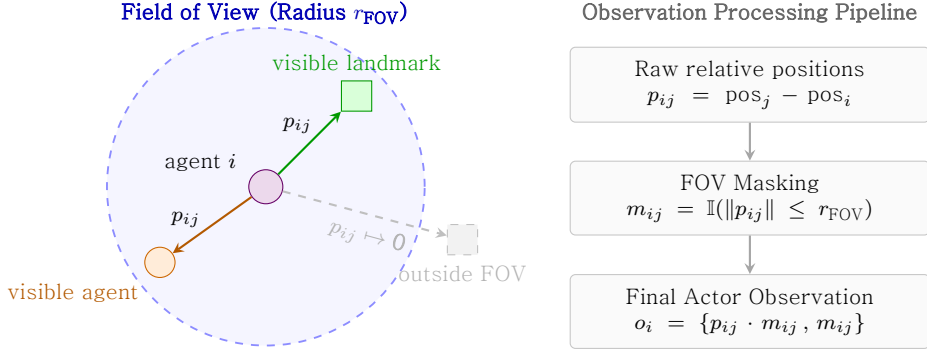
where H is the history length. The actor policy is decentralized because each agent-side input is built from local observations. During training, the critic follows the centralized-training decentralized-execution principle and uses the unmasked joint observation history.

2. FOV-Limited Observation Model

The original MPE `simple_spread` observation contains self velocity, self position, relative landmark positions, and relative positions of the other agents. To model partial observability, an FOV mask is applied to the actor observation. If a landmark or another agent lies outside the FOV radius, its relative position is set to zero. Visibility bits are appended to indicate which landmarks and agents are currently visible.

The masking operation is illustrated in Figure 3.2. The figure distinguishes the physical FOV constraint from the resulting actor observation vector, because this distinction is central to the actor-critic information mismatch used by MATDP. Objects inside the actor's field of view retain their relative-position vectors, while objects outside the FOV are masked to zero; the appended visibility bits allow the network to

distinguish masked objects from true zero-valued relative positions.



For an observed relative vector p_{ij} from agent i to object j , the masked vector is

$$\tilde{p}_{ij} = \begin{cases} p_{ij}, & \|p_{ij}\|_2 \leq r_{FOV}, \\ 0, & \|p_{ij}\|_2 > r_{FOV}, \end{cases} \quad (3.3)$$

where r_{FOV} is the FOV radius. This creates a direct mismatch between actor and critic inputs: the actor learns from masked local histories, while the critic estimates values from unmasked observations. This mismatch is intentional because it allows the critic to provide a stronger learning signal without giving unavailable information to the executed actor.

3. Actor Architecture

1) Per-Agent History Encoder

Each agent's observation history $\tau_t^i \in \mathbb{R}^{H \times D}$ is first projected into a hidden space. Positional encoding is added to preserve temporal order, and an agent embedding is added to identify which agent is being encoded.

The resulting sequence is processed by a Transformer encoder:

$$X_t^i = \text{Linear}(\tau_t^i) + PE + E_i, \quad (3.4)$$

$$h_t^i = \text{TransformerEncoder}(X_t^i)_H, \quad (3.5)$$

where $h_t^i \in \mathbb{R}^d$ is the final hidden state at the last history step. In the implementation, the main experiments use $H = 10$, hidden dimension $d = 128$, two Transformer layers, and four attention heads.

The Transformer encoder follows scaled dot-product self-attention [15]. For query, key, and value matrices Q , K , and V , the standard attention operation is

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V, \quad (3.6)$$

where d_k is the key dimension. This standard formulation is used as the sequence-modeling backbone. However, in the FOV-limited setting, not every object in the local observation is actually visible to each agent. The implementation first applies the FOV mask at the observation level by zeroing invisible relative-position vectors and appending visibility bits. Equivalently, the visibility-aware contribution of object tokens can be written as an additive mask inside the attention score:

$$\text{Attention}_{FOV}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + M_{FOV}\right)V, \quad (3.7)$$

with

$$M_{FOV}[i, j] = \begin{cases} 0, & \text{if object } j \text{ is visible to agent } i, \\ -\infty, & \text{if object } j \text{ is outside the FOV of agent } i. \end{cases} \quad (3.8)$$

This expression makes explicit the intended effect of the FOV preprocessing: visible objects may contribute to the agent representation, while outside-FOV objects cannot provide direct positional information to the actor. The appended visibility bits allow the network to distinguish a truly zero relative vector from an object that has been hidden by the FOV mask.

2) Team-Level Cross-Agent Encoder

The per-agent hidden states are stacked into a team representation:

$$H_t = [h_t^1, h_t^2, \dots, h_t^N] \in \mathbb{R}^{N \times d}. \quad (3.9)$$

A second Transformer encoder processes this agent sequence:

$$G_t = \text{TeamTransformer}(H_t). \quad (3.10)$$

The output G_t is flattened into a joint latent vector

$$g_t = \text{vec}(G_t) \in \mathbb{R}^{Nd}. \quad (3.11)$$

This team-level encoding lets the policy generate actions from a representation that includes both each agent's local history and cross-agent context.

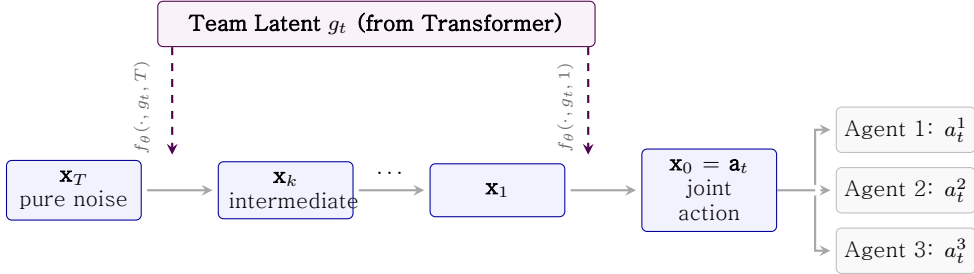
4. Joint Diffusion Policy

The MATDP actor uses a joint diffusion policy over the concatenated action vector:

$$\mathbf{a}_t = [a_t^1, a_t^2, \dots, a_t^N] \in \mathbb{R}^{NA}. \quad (3.12)$$

In the experimental environment, $N = 3$ and each agent has $A = 5$ action channels, so the joint action dimension is 15.

Figure 3.3 shows the resulting joint diffusion architecture. The team latent g_t conditions the denoising process, and the final denoised vector is then reshaped into the three agent actions. This makes the policy generate the full 15-dimensional joint action in one denoising chain instead of sampling three independent single-agent actions.



<Figure 3.3> Joint diffusion policy architecture

The MATDP actor uses a joint diffusion policy over the concatenated action vector $\mathbf{a}_t \in \mathbb{R}^{15}$. We perform $K = 10$ reverse denoising steps, where each transition evaluates a step log-probability required for PPO updates. In our experimental setup with $N = 3$ agents and $A = 5$ action channels per agent, the joint action is generated collectively, ensuring inter-agent dependencies are captured.

The forward diffusion process gradually adds Gaussian noise to a clean

action $\mathbf{a}_0$:

$$q(\mathbf{a}_k | \mathbf{a}_{k-1}) = \mathcal{N}(\mathbf{a}_k; \sqrt{1 - \beta_k} \mathbf{a}_{k-1}, \beta_k I), \quad (3.13)$$

and the closed form at step k is

$$q(\mathbf{a}_k | \mathbf{a}_0) = \mathcal{N}(\mathbf{a}_k; \sqrt{\bar{\alpha}_k} \mathbf{a}_0, (1 - \bar{\alpha}_k) I), \quad (3.14)$$

where $\alpha_k = 1 - \beta_k$ and $\bar{\alpha}_k = \prod_{j=1}^k \alpha_j$.

The reverse policy predicts the noise conditioned on the joint latent g_t and diffusion step embedding:

$$\hat{\epsilon}_\theta = f_\theta(\mathbf{a}_k, g_t, k). \quad (3.15)$$

Starting from Gaussian noise, the policy iteratively denoises the action for K steps and then applies a squashing function to keep actions inside the valid action range. The main experiments use $K = 10$ diffusion steps.

Generating the entire joint action simultaneously is an important design choice. Independent per-agent policies may choose individually reasonable movements that conflict at the team level. A joint diffusion policy can represent dependencies across agents because the denoising network observes the flattened team latent and predicts the full action vector together.

5. Centralized Critic

The critic uses unmasked observation histories from all agents. For a critic observation sequence $z_t \in \mathbb{R}^{N \times H \times D_c}$, all agent observations at each

history step are concatenated and passed through a Transformer encoder:

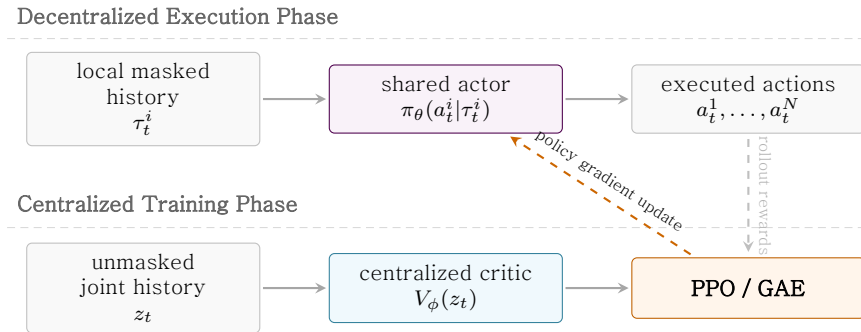
$$c_t = \text{CriticTransformer}(\text{Linear}(\text{Concat}(z_t^1, \dots, z_t^N))). \quad (3.16)$$

The final value estimate is produced by a value head after temporal mean pooling:

$$V_\phi(z_t) = \text{ValueHead}(\text{MeanPool}(c_t)). \quad (3.17)$$

This centralized critic reduces value-estimation difficulty during training, while the actor still learns to execute from masked local observations.

Figure 3.4 summarizes this centralized-training decentralized-execution structure. The important point is that the critic can use unmasked histories for learning, but the actor update must still produce a policy that acts from local FOV-masked histories. The dashed feedback path therefore represents gradient-based training information rather than extra observation information available during execution.



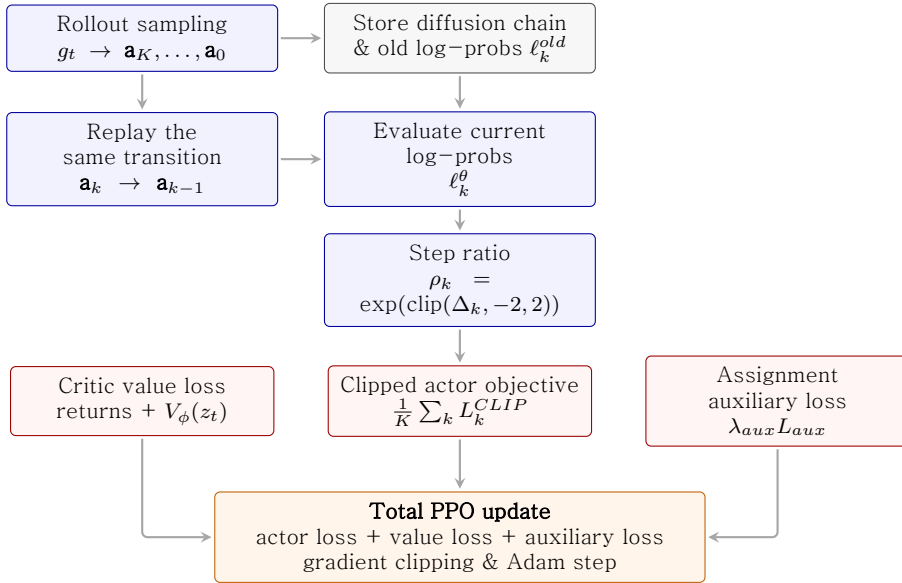
The actor never receives unmasked critic information at execution time.

<Figure 3.4> Centralized-training decentralized-execution structure

6. PPO Update for Diffusion Policy

PPO requires the log-probability ratio between the updated policy and the rollout policy. For a diffusion policy, this is computed from the denoising chain rather than from a single Gaussian distribution. During rollout, MATDP stores the sampled diffusion chain and the old step log probabilities. During update, the new policy evaluates the same denoising transitions.

Figure 3.5 shows how these stored rollout quantities enter the PPO update. The figure is placed before the equations because it identifies the three sources of the update objective: actor ratio terms, critic value loss, and assignment auxiliary loss.



<Figure 3.5> Step-wise PPO update for diffusion policy

For diffusion step k , the raw log-ratio is

$$\Delta_k = \log \pi_\theta(\mathbf{a}_{k-1}|\mathbf{a}_k, g_t) - \log \pi_{\theta_{old}}(\mathbf{a}_{k-1}|\mathbf{a}_k, g_t). \quad (3.18)$$

The implementation uses a step-average PPO objective. The ratio is clamped before exponentiation to avoid numerical instability:

$$\rho_k(\theta) = \exp(\text{clip}(\Delta_k, -2, 2)). \quad (3.19)$$

Unlike a standard Gaussian policy, where a single action log-probability is available for each rollout transition, the joint diffusion policy produces K denoising transitions and therefore K step-wise log-probabilities. The implementation stores these old step log-probabilities during rollout and recomputes the corresponding current log-probabilities during the PPO update. The resulting step-average diffusion ratio can be written as

$$\bar{\rho}_{diff}(\theta) = \frac{1}{K} \sum_{k=1}^K \exp(\text{clip}(\log \pi_{\theta}(\mathbf{a}_{k-1} | \mathbf{a}_k, g_t) - \log \pi_{\theta_{old}}(\mathbf{a}_{k-1} | \mathbf{a}_k, g_t), -2, 2)). \quad (3.20)$$

This averaging prevents a single denoising step from dominating the policy update. This is important because early reverse steps operate on high-noise intermediate actions, while later steps refine the structured joint action that is finally executed in the environment.

The clipped surrogate loss is

$$\mathcal{L}_{actor}(\theta) = -\frac{1}{K} \sum_{k=1}^K \mathbb{E} \left[\min \left(\rho_k(\theta) \hat{A}_t, \text{clip}(\rho_k(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right]. \quad (3.21)$$

Advantages are normalized within the batch:

$$\hat{A}_{norm} = \frac{\hat{A} - \mu_{\hat{A}}}{\sigma_{\hat{A}} + 10^{-8}}. \quad (3.22)$$

The critic is updated with a clipped value loss, and both actor and critic gradients are clipped for stability. Training also monitors approximate

KL divergence and stops overly large policy updates through a target-KL threshold.

7. Reward Shaping and Assignment Auxiliary Learning

1) Potential-Based Reward Shaping

The training reward includes the original team reward and shaping terms. The main shaping component is potential-based reward shaping (PBRS), which has the form

$$F(s_t, a_t, s_{t+1}) = \gamma \Phi(s_{t+1}) - \Phi(s_t). \quad (3.23)$$

According to the policy invariance theorem of Ng et al. [21], this form preserves the optimal policy when the potential depends only on state. In this thesis, the potential is based on the soft assignment cost between agents and landmarks:

$$\Phi(s) = - \sum_{i=1}^N \sum_{j=1}^N P_{ij}(s) d_{ij}(s), \quad (3.24)$$

where $d_{ij}(s)$ is the distance between agent i and landmark j , and $P_{ij}(s)$ is a soft assignment weight. The soft weights follow the idea of entropy-regularized assignment and Sinkhorn-style normalization [22]. The main flow uses the soft-assignment potential with scale 5.0.

In the implementation, the soft assignment matrix is computed from the pairwise distance matrix by an iterative Sinkhorn normalization

procedure. The unnormalized assignment score is initialized as

$$P_{ij}^{(0)} = \exp\left(-\frac{d_{ij}}{\tau}\right), \quad (3.25)$$

where τ is the assignment temperature. Each Sinkhorn iteration first normalizes rows,

$$\tilde{P}_{ij}^{(\ell)} = \frac{P_{ij}^{(\ell)}}{\sum_{j'=1}^N P_{ij'}^{(\ell)} + \epsilon}, \quad (3.26)$$

and then normalizes columns,

$$P_{ij}^{(\ell+1)} = \frac{\tilde{P}_{ij}^{(\ell)}}{\sum_{i'=1}^N \tilde{P}_{i'j}^{(\ell)} + \epsilon}. \quad (3.27)$$

After T iterations, $P(s) = P^{(T)}$ is used in the potential function. The temperature parameter controls the sharpness of the assignment: a smaller τ approaches a hard assignment, while a larger τ distributes mass more evenly across landmarks. The main experiments use $\tau = 0.2$ and $T = 30$ Sinkhorn iterations, implemented in log space for numerical stability.

The environment also provides a success coverage bonus and an FOV-local bonus. The coverage bonus rewards complete landmark coverage, while the FOV-local bonus gives a small signal for discovering visible landmarks or moving under no-visibility conditions. These shaping terms are used for practical learning stability, and their effect is evaluated through the controlled ablation and diagnostic experiments in Chapter 4.

2) Assignment Auxiliary Objective

In addition to PBRs, MATDP uses an assignment auxiliary objective. At each update, a target movement direction is computed from the

latest actor observation. The method builds a minimum-cost assignment between visible landmarks and agents using the Hungarian method [23]. For an assigned landmark with relative vector v_i , the target movement is

$$m_i^{target} = \text{clip}(v_i, -1, 1). \quad (3.28)$$

The actor's deterministic action is converted into an effective movement vector, and a mean-squared error loss is applied:

$$\mathcal{L}_{aux} = \frac{1}{N} \sum_{i=1}^N \|m_i^{det} - m_i^{target}\|_2^2. \quad (3.29)$$

The final actor loss is

$$\mathcal{L}_{total} = \mathcal{L}_{actor} + \lambda_{aux} \mathcal{L}_{aux}, \quad (3.30)$$

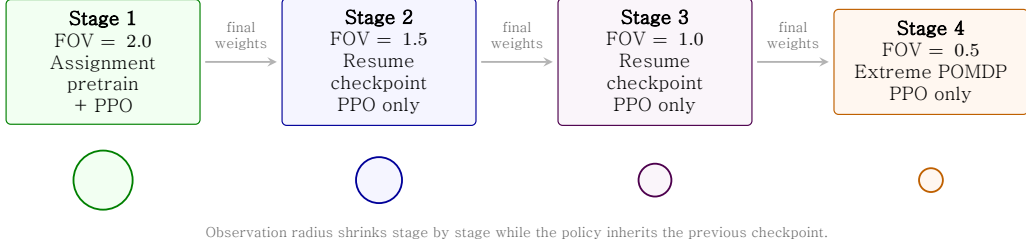
where the main curriculum uses $\lambda_{aux} = 10$. This auxiliary objective is also used for the FOV=2.0 assignment pretraining stage. It does not replace the learned policy during execution; it only provides a stabilizing direction during training.

8. Curriculum Training Procedure

The main training protocol uses a four-stage FOV flow. Stage 1 starts with FOV=2.0 and includes 1000 steps of assignment pretraining. Stages 2, 3, and 4 use FOV=1.5, FOV=1.0, and FOV=0.5, respectively. Each later stage resumes the final checkpoint from the previous stage and performs online PPO training without additional assignment pretraining.

Figure 3.6 summarizes this training flow. It makes explicit that the

observation radius shrinks stage by stage while the policy inherits the previous checkpoint. Thus, the later stages test adaptation under stronger partial observability rather than training each FOV setting from scratch.



<Figure 3.6> Four-stage FOV flow

This flow is designed to avoid training directly from the hardest partial-observability setting. The policy first learns coordination under easier visibility, then adapts to increasingly limited observations. The next chapter evaluates whether this flow helps preserve success as the FOV becomes smaller.

9. Chapter Summary

The proposed MATDP framework combines Transformer-based observation-history modeling, team-level cross-agent encoding, joint diffusion action generation, PPO optimization, centralized value estimation, PBRS, and assignment auxiliary stabilization. These components are designed to address the three main difficulties of the target problem: partial observability, coordinated joint action generation, and unstable online optimization of an expressive diffusion policy.

IV. Experimental Evaluation and Analysis

The experiments in this chapter are organized as an evidence ladder. First, a controlled four-stage FOV flow compares MATDP with two internal variants, MATD and MATG, under the same schedule and training budget. Second, a fixed-FOV MAPPO baseline checks whether the proposed architecture remains competitive when both algorithms are evaluated with the same PettingZoo environment reward and the MATDP auxiliary reward terms are disabled. Third, diagnostic ablations and parameter-sensitivity runs are used to identify the boundary conditions of the method under extreme partial observability.

The central finding is intentionally bounded. MATDP achieves the strongest `success_last100` profile in the internal flow comparison, and it improves the reward profile while lowering the maximum landmark-to-nearest-agent distance in the aligned MAPPO comparison. However, this does not mean that the FOV=0.5 setting is solved. In the hardest condition, deterministic evaluation remains weak, so the final policy should be interpreted as a stochastic and partially reliable coordination policy rather than a fully robust deterministic solution.

1. Experimental Environment and Setup

The experiments are conducted in the MPE `simple_spread` environment [6, 24]. The task contains three agents and three landmarks. Each agent receives an observation of the local scene and outputs a five-channel action vector corresponding to the MPE

movement interface, `[noop,left,right,down,up]`. Since the proposed policy generates a centralized joint action for all three agents, the joint action dimension is 15. The agents are trained to occupy different landmarks while avoiding inefficient overlap, so the task tests both continuous control and multi-agent coordination.

To model partial observability, a field-of-view (FOV) mask is applied to the actor observation. A landmark or another agent outside the FOV radius is hidden from the policy input. This setting is used as an abstraction of blind spots in autonomous driving: the vehicle may not directly observe all relevant objects, but it still needs to act based on short observation history and learned coordination priors. The critic follows the centralized-training decentralized-execution principle and uses unmasked observations during training. Thus, the actor remains partially observable, while the critic can provide a more informative value estimate.

The main training protocol is a four-stage flow. The first stage uses $\text{FOV}=2.0$, which is close to a near-full observation setting. The following stages gradually reduce the observation radius to $\text{FOV}=1.5$, $\text{FOV}=1.0$, and $\text{FOV}=0.5$. This design avoids starting directly from an extremely sparse observation regime and tests whether a policy learned under easier visibility can be adapted to stronger partial observability. The $\text{FOV}=2.0$ stage uses assignment pretraining for 1000 supervised steps. $\text{FOV}=1.5$ – $\text{FOV}=0.5$ do not use additional pretraining and instead resume the final checkpoint from the previous stage.

The experiments were run on a workstation equipped with two NVIDIA RTX A5000 GPUs. Each flow stage is trained for 1000 episodes with a maximum episode length of 50 steps. Unless otherwise

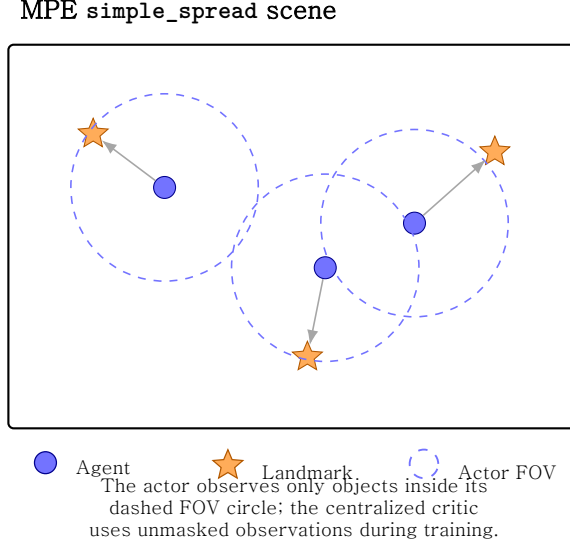
stated, PBRS is enabled with a soft-assignment potential, the actor uses FOV-masked observations, and the critic uses unmasked observations. All reported success statistics follow the same distance-based success rule used by the experiment logs, with a success threshold of 0.2. For the fixed-FOV MAPPO baseline reported later, MATDP is rerun with auxiliary reward terms disabled, and both algorithms are evaluated with the original PettingZoo environment reward.

Table 4.1 lists the main settings used by the MATDP, MATD, and MATG flow experiments. These values define the common training budget and model configuration for the internal comparison, including the number of agents, action dimension, FOV schedule, PPO settings, diffusion settings, and stabilization terms.

<Table 4.1> Main hyperparameter settings

Hyperparameter	Value
Environment	MPE <code>simple_spread</code>
Number of agents / landmarks	3 / 3
Single-agent action dimension	5
Joint action dimension	15
Episodes per stage	1000
Max steps per episode	50
Flow FOV values	2.0, 1.5, 1.0, 0.5
Assignment pretraining	1000 steps in FOV=2.0 only
Assignment auxiliary	Enabled, movement target, weight 10
Actor learning rate	2×10^{-5}
Critic learning rate	3×10^{-4}
PPO target KL	0.08
Batch size	128
PPO epochs	5
Diffusion steps	10
Diffusion std scale	0.3
Diffusion PPO objective	Step-average log probability
PBRS	Enabled, soft-assignment potential
Coverage / FOV local bonus	Enabled
Critic observation source	Unmasked observation
Belief auxiliary / actor memory	Disabled in main flow

Figure 4.1 illustrates the FOV-limited `simple_spread` setup used in the experiments. The dashed circles represent the actor-side visibility constraint, while the centralized critic is trained with unmasked observations.



<Figure 4.1> FOV-limited MPE `simple_spread` environment

With these settings, the experiment progressively increases partial observability while keeping model and training parameters fixed, so performance changes can be attributed mainly to the FOV flow.

2. Compared Methods and Evaluation Metrics

Three methods are compared in the main experiments. MATDP is the proposed full model. It combines a Transformer encoder for observation-history modeling, a 15-dimensional joint diffusion policy for multi-agent action generation, and PPO for online policy optimization. The diffusion policy models the joint action of all agents at once, so the action generator is explicitly aware of cross-agent coordination.

MATD keeps the Transformer and joint diffusion policy but removes online PPO updates during the environment rollout. It is still allowed to use the FOV=2.0 assignment pretraining stage. This ablation tests whether the learned diffusion prior alone is enough, or whether online reinforcement learning is necessary for adaptation under changing FOV conditions.

MATG replaces the diffusion policy with a standard Gaussian policy while keeping the Transformer encoder and PPO. This baseline tests whether the diffusion action generator provides benefits beyond a simpler continuous-control policy under the same CTDE, PBRS, and flow setup.

Table 4.2 summarizes these architectural differences. The table is included to make clear that MATD removes online PPO adaptation, whereas MATG keeps PPO but replaces the diffusion actor.

◁Table 4.2▷ Architecture comparison of the evaluated methods

Method	Transformer	Diffusion policy	PPO update	Joint action
MATDP	Yes	Yes	Yes	15D joint
MATD	Yes	Yes	No	15D joint
MATG	Yes	No	Yes	Gaussian action policy

MAPPO is used as an external baseline in a separate fixed-FOV comparison rather than as a fourth row in the main flow tables. This separation is necessary because the MAPPO runs use an aligned PettingZoo environment-reward protocol, while the main flow results evaluate MATDP, MATD, and MATG under the staged flow with the training stabilizers described above. The MAPPO comparison therefore answers a different question from the internal ablation: it tests whether MATDP remains competitive against a mature CTDE baseline after auxiliary reward terms are removed.

The primary metric is `success_last100`, the mean success rate

over the final 100 training episodes. This metric reflects late-stage performance after the policy has adapted within each FOV condition. `total_success` counts successful episodes over the full run and is used as a secondary stability indicator. `coverage_last100` measures the fraction of landmarks covered by agents. `maxdist_last100` is the average maximum landmark-to-nearest-agent distance; lower values mean that even the least-covered landmark is closer to some agent. `reward_last100` summarizes the final reward level. Finally, `det_eval_last` reports the last deterministic evaluation success rate, where stochastic sampling is disabled or reduced.

Together, the three-method comparison separates the contributions of online PPO adaptation and the diffusion action generator, while the later MAPPO experiment provides an external baseline under a stricter fixed-FOV reward protocol.

3. Main flow Results

Table 4.3 reports the first step of the evidence ladder: a controlled flow comparison in which MATDP, MATD, and MATG share the same FOV schedule and training budget. The most important observation is that MATDP achieves the highest `success_last100` in every flow stage. This pattern holds from the near-full observation setting FOV=2.0 to the most difficult FOV=0.5.

<Table 4.3> Flow success rate comparison

FOV	MATDP	MATD	MATG
2.0	0.190	0.060	0.070
1.5	0.170	0.060	0.080
1.0	0.150	0.050	0.100
0.5	0.150	0.010	0.110

The comparison between MATDP and MATD isolates the role of online PPO adaptation. MATD keeps the same Transformer and diffusion policy but removes online policy updates during the environment rollout. In FOV=0.5, MATD reaches only 0.010 success in the final 100 episodes, while MATDP reaches 0.150. The total number of successful episodes shows the same pattern: MATDP obtains 81 successful episodes in FOV=0.5, while MATD obtains only 12. These results indicate that assignment pretraining alone is not sufficient for adapting the policy to the decreasing FOV conditions. Online PPO updates are needed to preserve useful behavior as the observation radius becomes smaller.

The comparison between MATDP and MATG isolates the contribution of the joint diffusion action generator. MATG is a strong internal baseline because Gaussian PPO is easier to optimize than a diffusion policy and benefits from the same Transformer encoder, CTDE critic, PBRS shaping, and flow schedule. Nevertheless, MATDP still obtains a higher success rate in all four flow stages. In FOV=1.0, MATDP reaches 0.150 success compared with 0.100 for MATG; in FOV=0.5, MATDP reaches 0.150 compared with 0.110 for MATG. This supports the claim that joint diffusion action modeling improves the main success metric. At the same time, MATG remains competitive on some auxiliary indicators. For example, in FOV=1.0, MATG obtains higher coverage and a higher deterministic evaluation score. Therefore, the conclusion is not that MATDP is superior on every metric, but that it gives the best success-rate profile across the flow.

Table 4.4 reports these secondary indicators. It is used to qualify the success-rate conclusion rather than to replace it: MATDP has the strongest main success profile, while some auxiliary metrics remain

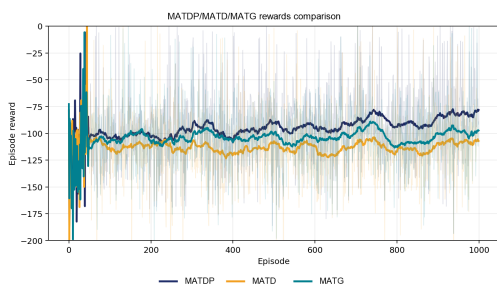
mixed.

<Table 4.4> Auxiliary flow metrics

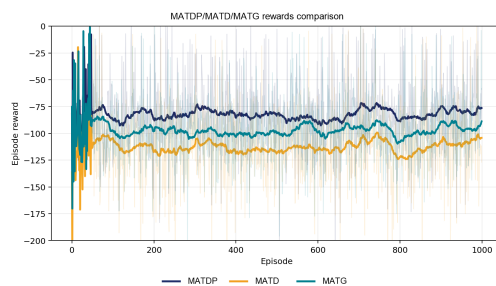
FOV	Method	Total success	Coverage	MaxDist	Reward	DetEval
2.0	MATDP	101	0.290	0.368	-80.85	0.100
2.0	MATD	47	0.293	0.420	-108.46	0.100
2.0	MATG	46	0.283	0.425	-97.88	0.000
1.5	MATDP	139	0.300	0.358	-78.38	0.000
1.5	MATD	40	0.290	0.431	-108.15	0.000
1.5	MATG	64	0.290	0.398	-92.63	0.100
1.0	MATDP	143	0.293	0.430	-80.82	0.200
1.0	MATD	39	0.290	0.507	-121.62	0.000
1.0	MATG	87	0.330	0.465	-92.74	0.300
0.5	MATDP	81	0.293	0.636	-105.61	0.000
0.5	MATD	12	0.210	0.596	-143.92	0.000
0.5	MATG	61	0.300	0.598	-109.86	0.000

Figure 4.2 provides a complementary view of the training dynamics. Across the four FOV stages, MATDP usually maintains a higher smoothed episode reward than MATD and remains close to or above MATG. In each panel, the lighter traces show raw episode rewards, while the darker curves show smoothed trends. The curves should be interpreted as diagnostic evidence for reward progression rather than as a formal variance-based stability test.

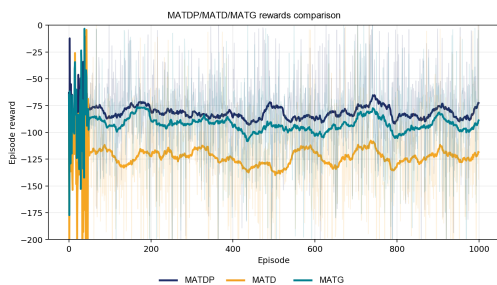
MATDP is also more robust across FOV stages than expected. Its success decreases from 0.190 in FOV=2.0 to 0.150 in FOV=0.5, while MATD drops from 0.060 to 0.010. This suggests that online adaptation helps preserve the diffusion policy under stronger partial observability. However, the deterministic evaluation result in FOV=0.5 is still 0.000, and the final reward has larger variance than in the easier stages. This indicates that the FOV=0.5 successes are not yet produced by a fully stable deterministic policy. The stage-wise success trend is visualized in Figure 4.3.



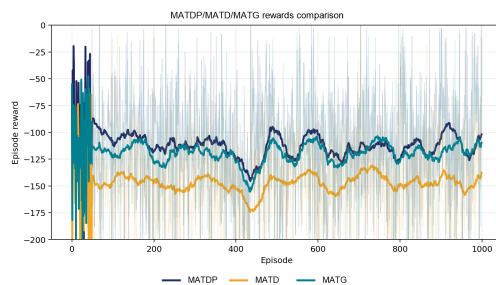
(a) FOV=2.0



(b) FOV=1.5

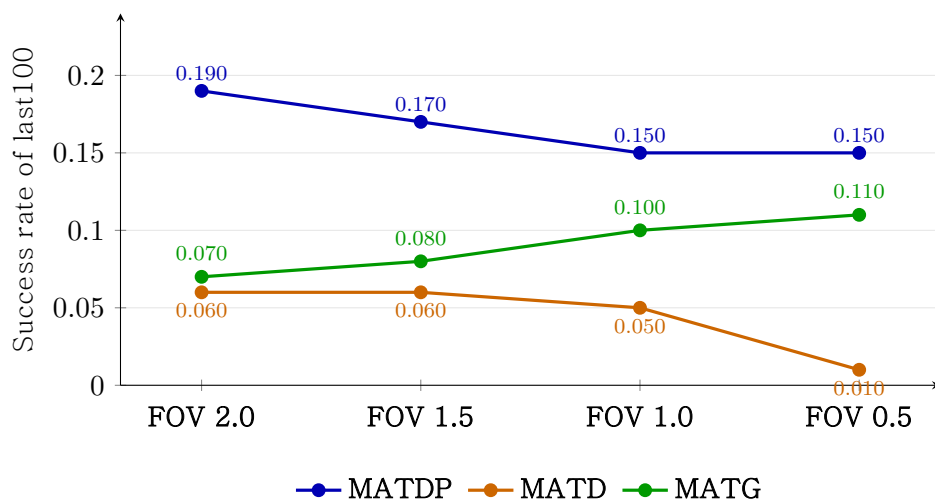


(c) FOV=1.0



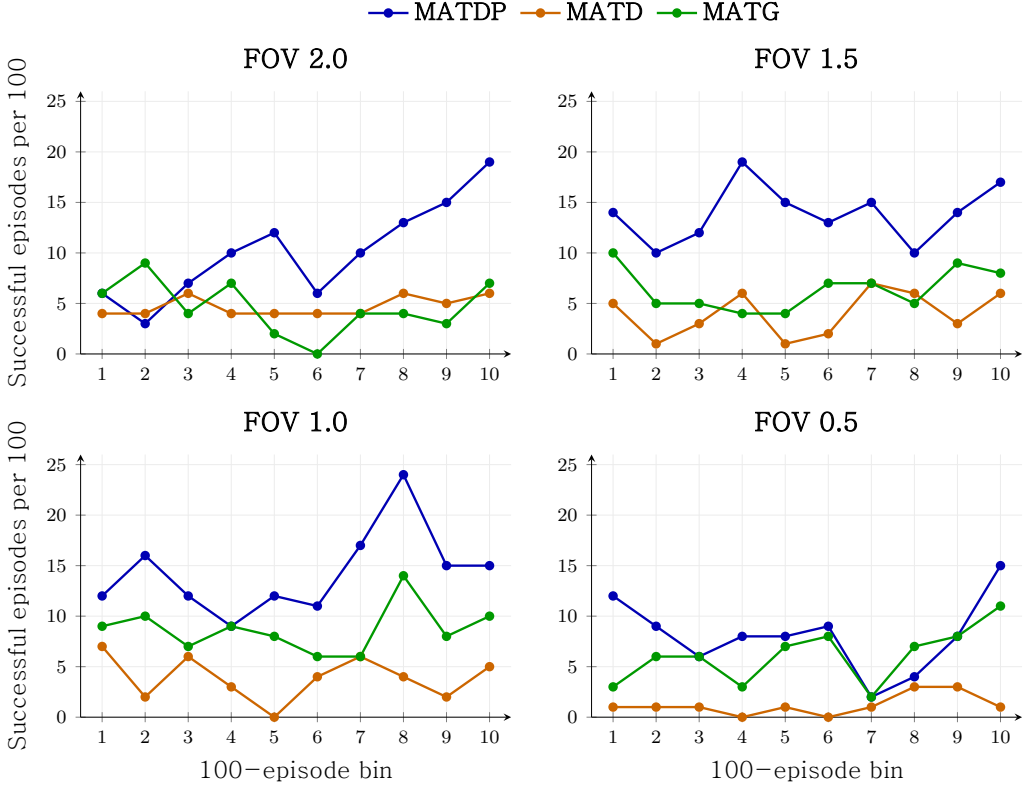
(d) FOV=0.5

<Figure 4.2> Reward trajectories across FOV flow



<Figure 4.3> Flow success rate across FOV stages

To check whether the final success rate is concentrated in a single short window or distributed across training, Figure 4.4 reports the successful episodes in each 100-episode bin. The four panels correspond to the four FOV stages from FOV=2.0 to FOV=0.5, so the figure also shows how success counts change as visibility becomes more restrictive.



<Figure 4.4> Per-100-episode success counts

For MATDP, the FOV=2.0 per-100 success counts are [6, 3, 7, 10, 12, 6, 10, 13, 15, 19], and the FOV=0.5 counts are [12, 9, 6, 8, 8, 9, 2, 4, 8, 15]. These trends show that MATDP does not simply succeed in the first stage and then collapse. It continues to produce successful episodes even under the smallest FOV, although the behavior remains stochastic.

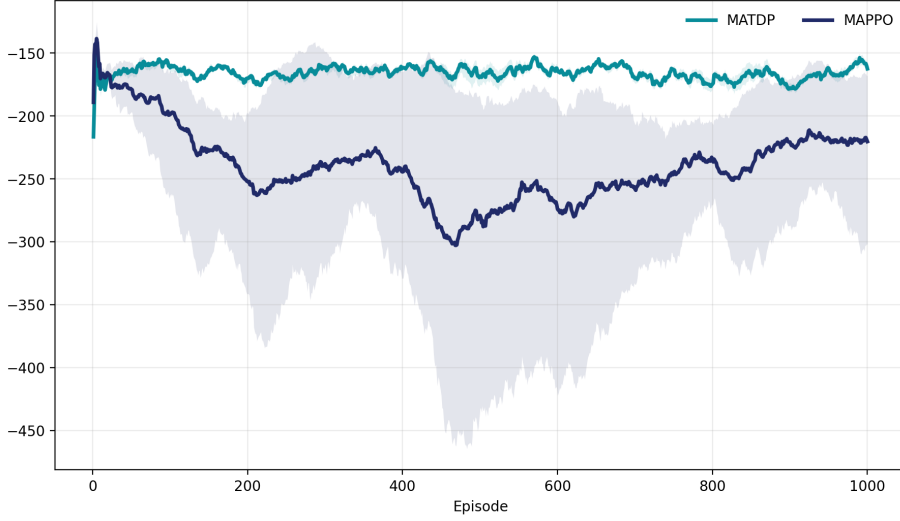
The main flow result is therefore two-sided: MATDP has the

best success-rate profile among the internal variants, but the FOV=0.5 deterministic result shows that extreme partial observability is still not fully solved. The next question is whether this advantage remains meaningful when MATDP is compared with a mature external MARL baseline under a stricter fixed-FOV and environment-reward protocol. The following section addresses this question using MAPPO.

4. MATDP-MAPPO Fixed-FOV Baseline Comparison

To make the external baseline comparison more conservative, a separate fixed-FOV experiment compares MATDP with MAPPO under the same PettingZoo `simple_spread` reward protocol. In this comparison, MATDP is evaluated with the auxiliary reward terms disabled: PBRS, the assignment auxiliary, the belief auxiliary, the coverage bonus, and the FOV-local bonus are not used. The reported reward is therefore the original environment reward, aligned with the MAPPO PettingZoo training setup. The four fixed FOV settings are FOV=2.0, 1.5, 1.0, and 0.5, and each run uses 1000 episodes with a maximum episode length of 50 steps. This comparison is not mixed into the main flow tables because it uses fixed-FOV runs rather than the staged flow protocol.

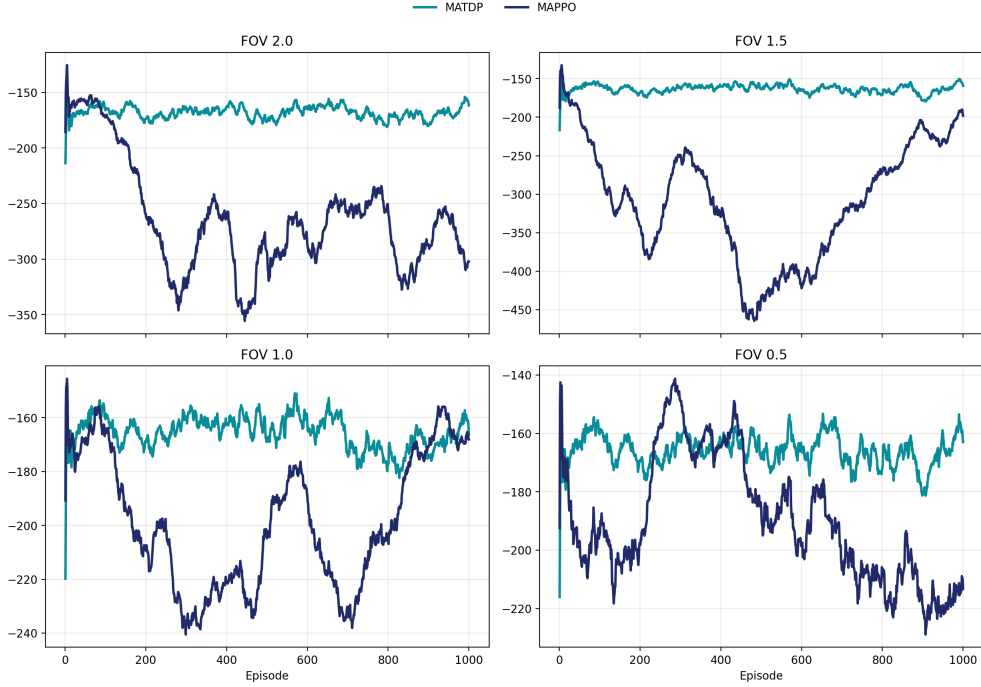
Figure 4.5 summarizes the average reward trend across the four fixed-FOV settings. The curves show rolling-50 environment reward averaged over FOV=2.0, 1.5, 1.0, and 0.5, and the shaded region shows the range across FOV settings. MATDP maintains a higher rolling environment reward than MAPPO for most of training under the aligned reward protocol. This supports the reward-level advantage of the MATDP policy architecture when the auxiliary reward terms are removed.



<Figure 4.5> Fixed-FOV MATDP-MAPPO reward trend

The per-FOV curves in Figure 4.6 show where this reward advantage appears. Each panel plots the rolling-50 environment reward for one fixed FOV setting under the aligned PettingZoo protocol. In the last 100 episodes, MATDP obtains higher environment reward than MAPPO at FOV=2.0 (-162.88 vs. -282.44), FOV=1.5 (-159.70 vs. -213.59), and FOV=0.5 (-161.35 vs. -216.06). The FOV=1.0 case is the exception: MAPPO is slightly higher than MATDP (-163.90 vs. -165.18). The correct conclusion is therefore not that MATDP is higher in every fixed-FOV condition, but that it gives a stronger reward profile in three of the four aligned MAPPO comparisons.

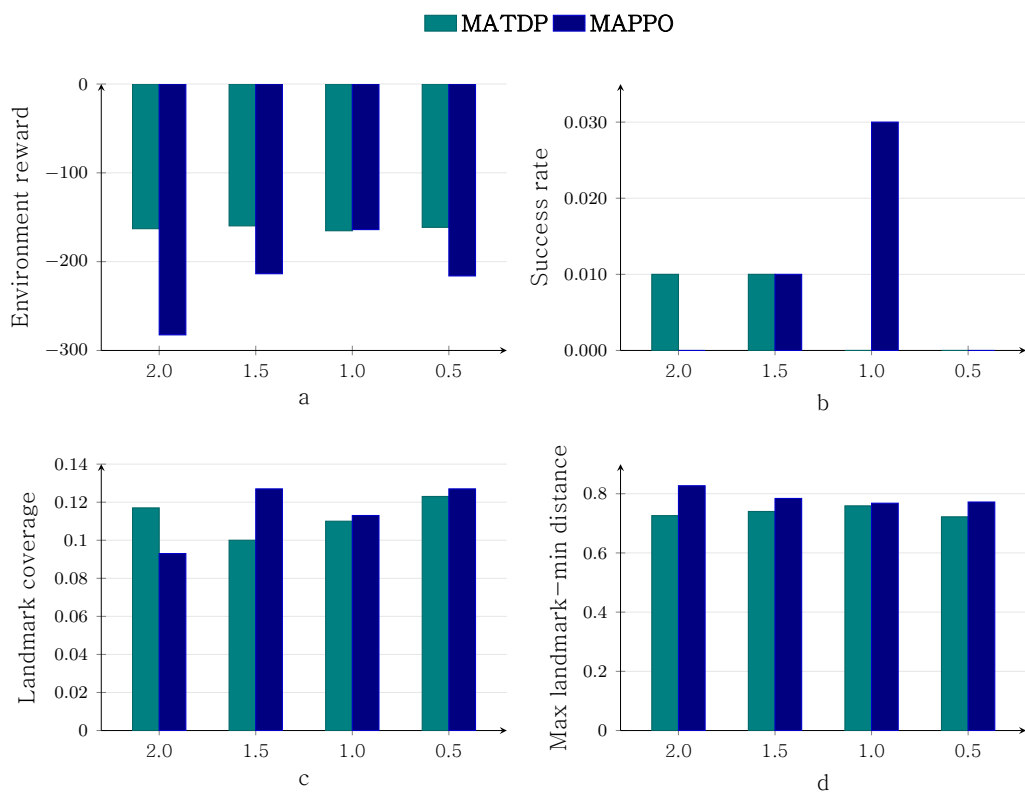
Figure 4.7 compares the final 100 episodes using reward, success rate, landmark coverage, and maximum landmark-to-nearest-agent distance. Higher values are better for reward, success rate, and landmark coverage, while lower values are better for the maximum landmark-to-nearest-agent distance. The success results remain sparse for both algorithms. Across all 1000 episodes, the success counts



⟨Figure 4.6⟩ Per-FOV MATDP-MAPPO reward trends

for MATDP/MAPPO are 4/4 at FOV=2.0, 4/3 at FOV=1.5, 3/6 at FOV=1.0, and 2/4 at FOV=0.5. The last-100 success rates are also mixed, with MAPPO higher at FOV=1.0. Coverage is similarly mixed across FOV settings. In contrast, the maximum landmark-to-nearest-agent distance is lower for MATDP in all four fixed-FOV settings. This means that the least-covered landmark is closer to some agent under MATDP, even when successful episodes are rare.

Overall, the fixed-FOV MAPPO baseline strengthens the fairness of the empirical comparison, but it also narrows the claim. MATDP shows a clear reward advantage and lower maximum landmark-to-nearest-agent distance in this aligned setting, while MAPPO remains competitive on success-related indicators in some FOV conditions. The result should therefore be read as evidence for improved reward stability and spatial



<Figure 4.7> Last-100 MATDP-MAPPO metric comparison

coverage under aligned rewards, not as a general success-rate advantage over MAPPO.

The mechanism-level interpretation is consistent with the design of the two policies. MAPPO follows the CTDE principle, but the actor still has to choose actions from FOV-masked observations during execution. When landmarks or other agents are outside the observation radius, the critic can provide a centralized value estimate during training, but the actor cannot directly reconstruct the missing local information at decision time. In `simple_spread`, the success condition is also strict because all landmarks must be covered simultaneously. As a result, successful episodes are sparse, and the advantage signal can become noisy even when the reward and distance metrics improve.

MATDP addresses this setting through a different trade-off. The Transformer history encoder gives the actor a short temporal context, while the joint diffusion policy samples a full 15-dimensional team action conditioned on the team latent representation. This design is computationally heavier than a Gaussian policy, but it gives the policy a more structured way to model correlated agent movements. The observed reward improvement and lower maximum landmark-to-nearest-agent distance are therefore consistent with better spatial coordination. At the same time, the sparse success rates show that this coordination prior does not fully solve the hardest partial-observation cases.

5. Ablation Studies

1) No-pretraining Baseline at FOV=2.0

The first ablation removes assignment pretraining and evaluates the three methods directly at FOV=2.0. This setting is less difficult than the full flow because the observation radius is large, but it is useful for checking whether the models can learn without a supervised warm start. The results come from a separate no-pretraining experiment conducted under the FOV=2.0 setting.

Table 4.5 reports this no-pretraining comparison. The table is separated from the main flow results because it uses a different training protocol and is intended as a diagnostic ablation.

<Table 4.5> No-pretraining comparison at FOV=2.0

Method	Success	Total	Coverage	MaxDist	Reward	DetEval
MATDP	0.100	69	0.250	0.415	-91.39	0.400
MATD	0.010	3	0.130	0.696	-141.51	0.000
MATG	0.050	48	0.303	0.438	-102.43	0.200

Without pretraining, MATDP still obtains the highest success rate, total success count, reward, and deterministic evaluation score. The deterministic result is particularly important: MATDP reaches 0.400, while MATG reaches 0.200 and MATD remains at 0.000. This suggests that, in the easier FOV=2.0 setting, the proposed model does not only benefit from stochastic exploration; it also learns a more deterministic policy than the MATG baseline. MATG has higher coverage, which means it can spread agents over the map, but this does not translate into the same level of successful task completion.

2) Actor–side Memory Diagnostics

The second ablation studies whether explicit actor–side memory can help in the extreme FOV=0.5 condition. These diagnostic runs were conducted separately from the main flow because the memory variants use a different training configuration. The comparison uses longer 75–step episodes and 1500 FOV=0.5 pretraining steps, so these numbers should not be mixed with the main flow table as a direct model comparison. Their purpose is to analyze the observation bottleneck.

Table 4.6 summarizes the actor–side memory diagnostics. The key comparison is not which variant wins overall, but whether additional memory channels meaningfully repair the extreme FOV observation bottleneck.

<Table 4.6> Actor–side memory diagnostics in FOV=0.5

Method	Success	Total	Coverage	MaxDist	Visible fraction
No memory	0.010	8	0.210	0.718	0.036
Bit memory	0.030	13	0.263	0.672	0.042
Structured memory	0.020	10	0.290	0.707	0.038

The visible fraction is low in all three runs, which confirms that FOV=0.5 creates a genuine observation bottleneck. Bit memory gives the highest success rate and total success count. Structured memory gives the highest coverage, but it does not produce the highest success rate. This suggests that adding more memory channels can help spatial exploration, but the policy still needs enough samples to learn how to use position and age information reliably. In this task, simple memory bits are easier to exploit than a higher–dimensional structured memory vector.

These ablations support two points: MATDP can learn without assignment pretraining under large FOV, and explicit memory gives only

weak improvement under extreme FOV.

6. Extreme POMDP Analysis

The FOV=0.5 stage is the clearest boundary condition of the proposed framework. Although the main flow result gives MATDP a final stochastic success rate of 0.150, the deterministic evaluation remains 0.000. This means that FOV=0.5 should not be presented as a solved setting. Instead, it should be interpreted as an extreme POMDP condition where stochastic exploration, learned coordination priors, and reward shaping occasionally produce successful episodes, but the actor does not yet receive enough reliable information to reproduce the behavior deterministically.

To test whether the failure can be repaired by tuning PPO and exploration parameters, a separate FOV=0.5 parameter sensitivity analysis was conducted. The P0 setting is a conservative FOV=0.5 baseline from the earlier pre1000 rescue setup. P1 relaxes PPO by increasing the actor learning rate and target KL. P2+ increases diffusion exploration and uses a longer 50-step episode with a larger batch size. These runs are diagnostic and are not part of the main flow comparison.

Table 4.7 reports the parameter-sensitivity diagnostics. These runs test whether stronger updates or exploration can solve the FOV=0.5 failure without changing the information available to the actor.

<Table 4.7> FOV=0.5 parameter sensitivity diagnostics

Config	Success	Total	Coverage	MaxDist	DetEval	Approx KL
P0 baseline	0.000	12	0.180	0.688	0.100	0.003
P1 relaxed PPO	0.000	6	0.163	0.713	0.100	0.009
P2+ stronger exploration	0.000	11	0.200	0.695	0.000	0.006

The rescue experiments show that parameter tuning alone is not a

sufficient solution. P1 increases the policy update magnitude, as shown by the higher approximate KL, but it does not improve success. P2+ improves coverage to 0.200, but the final success rate remains 0.000 and deterministic evaluation drops to 0.000. This suggests that stronger exploration can move agents through more of the environment, but PPO still fails to consolidate rare successful trajectories into a stable policy when the actor's observations are too sparse.

This behavior can be explained by the perception–action loop. In $\text{FOV}=0.5$, agents often act without seeing any landmark. If a random exploratory movement later produces a better global configuration, the local actor observation may not contain enough information to explain why the action was good. The CTDE critic can estimate value from unmasked information, but the actor still has to produce actions from masked observations. When useful observations are rare, the Transformer receives too few meaningful memory events, and the diffusion policy receives sparse and noisy gradient signals due to poor credit assignment under limited observability. This is why memory bits and longer episodes help slightly but do not solve the problem.

From an autonomous–driving perspective, this is equivalent to a vehicle entering a scenario with dense occlusions without communication from other vehicles or infrastructure. Local perception alone may occasionally lead to safe behavior, but stable coordination requires either richer memory, explicit communication, or stronger offline priors. Therefore, $\text{FOV}=0.5$ is best treated as an information–flow boundary of the proposed framework rather than as a solved benchmark.

The extreme–POMDP diagnostics therefore expose a structural limitation: the policy needs better information flow, not only more

PPO updates or stronger random exploration. This boundary condition also identifies a concrete direction for future MARL communication mechanisms: they should supply the missing team-level information that the FOV-masked actor cannot infer from local history alone.

7. Limitations of MATDP Observed in the Experiments

The advantage of MATDP should be interpreted together with its trade-offs and boundary conditions. The first trade-off is computation for performance. The main flow shows that MATDP achieves the best `success_last100` profile among the compared methods, but this advantage is not free. Compared with MATG, MATDP uses a joint diffusion policy with $K = 10$ reverse denoising steps for each sampled joint action. During PPO updates, the method also stores the denoising chain and recomputes step-wise log probabilities for the step-average diffusion ratio. Therefore, MATDP is computationally heavier than the MATG baseline in both training and action generation. This thesis does not report a separate wall-clock benchmark, so the claim is limited to model and update complexity rather than measured runtime.

A second trade-off is optimization complexity. MATDP depends on several interacting design choices, including the diffusion noise scale, the target KL threshold, gradient clipping, the step-average PPO ratio, and the balance between reinforcement learning, PBRs, and the assignment auxiliary loss. Some of these stabilizers are also used by MATD and MATG, so they should not be treated as a weakness unique to MATDP. The MATDP-specific issue is that the diffusion actor makes the training pipeline more sensitive and less transparent than a simpler Gaussian

policy. In practice, this means that MATDP may require more careful tuning before it can be transferred to another task.

A third trade-off is coordination expressiveness versus scalability. In the present task, the joint action dimension is only 15 because there are three agents and each agent has five action channels. If the number of agents increases, the joint diffusion policy must model a higher-dimensional action vector and a larger team-level latent representation. This design is useful for representing cross-agent dependencies, but it may become expensive or difficult to optimize in larger teams without factorization, parameter sharing, or hierarchical action generation.

Finally, the most important boundary condition is $FOV=0.5$. MATDP still obtains stochastic successes in the main flow, but the deterministic evaluation remains 0.000. This result means that the policy has not learned a reliably reproducible coordination strategy under extreme partial observability. The failure is not simply a lack of stronger PPO updates, because the parameter-sensitivity diagnostics do not repair it. Instead, the evidence suggests that MATDP reaches its current boundary when the actor's local history contains too few useful observations. Under such conditions, a more expressive action generator cannot replace missing information.

The fixed-FOV MAPPO comparison reinforces this bounded interpretation. Under the aligned environment-reward protocol, MATDP improves reward in three of four FOV settings and lowers the final maximum landmark-to-nearest-agent distance in all four settings. However, the success rates remain low for both algorithms, and MAPPO is competitive or better on the $FOV=1.0$ success indicators.

Therefore, the external-baseline claim should remain limited to reward and spatial-coverage trends rather than general MAPPO dominance.

8. Chapter Summary

The MPE `simple_spread` evaluation used a four-stage FOV flow and three controlled methods: MATDP, MATD, and MATG. All main flow results were obtained under the same flow schedule and training budget. This internal comparison shows that MATDP achieves the highest `success_last100` across all four FOV stages. The MATDP-MATD comparison demonstrates the importance of online PPO adaptation, while the MATDP-MATG comparison supports the contribution of the joint diffusion policy to the main success-rate metric.

The fixed-FOV MATDP-MAPPO comparison provides an external baseline under an aligned PettingZoo environment-reward protocol. In this setting, MATDP has a stronger final reward profile in three of the four FOV settings and a lower maximum landmark-to-nearest-agent distance in all four settings. The success results remain sparse and mixed, so the MAPPO comparison supports a bounded reward-and-coverage claim rather than a general claim that MATDP is superior on every metric.

The hardest FOV=0.5 condition remains unresolved in a deterministic sense. Although MATDP obtains the best stochastic success rate in the flow, deterministic evaluation at FOV=0.5 is still 0.000. The memory and parameter-sensitivity diagnostics show that stronger exploration, relaxed PPO settings, and simple actor-side memory provide only weak improvements. The main bottleneck is therefore information flow: the actor cannot reliably infer missing landmark and team information from

FOV-masked local history alone.

These results define the practical trade-offs of MATDP. The joint diffusion actor improves the success profile and spatial-distance behavior, but it also increases computation, introduces additional optimization sensitivity, and may face scalability issues as the joint action dimension grows. The findings motivate future work on explicit communication, more stable memory mechanisms, stronger offline pretraining, and more efficient factorized joint-action modeling for extreme partially observable multi-agent control.

V. Conclusion and Future Work

1. Conclusion

This thesis asked whether a Transformer–Diffusion–PPO actor can improve cooperative continuous control when each agent observes only a limited local field of view. The answer from the experiments is positive, but bounded. In the controlled MPE `simple_spread` flow, MATDP achieves the best `success_last100` among the compared methods at all four FOV stages. At the same time, the hardest FOV=0.5 setting is not solved in a deterministic sense, which shows that the main remaining bottleneck is information availability rather than only optimizer strength.

The implemented framework combines three choices that are useful together. First, each actor receives a short FOV–masked observation history, and a Transformer encoder converts this history into a temporal latent representation. A second Transformer then processes the agent embeddings before action generation. Second, the policy generates the 15–dimensional joint action for all three agents through a diffusion denoising process, so cross–agent action dependencies are represented directly instead of being factorized into independent per–agent policies. Third, PPO is adapted to this diffusion actor through step–average log–probability ratios, while the centralized critic, advantage normalization, gradient clipping, PBRS, and assignment auxiliary learning keep online updates stable enough for flow training.

The main experimental comparison gives the clearest evidence for the method. Across the controlled four–stage flow, MATDP reaches

`success_last100` values of 0.190, 0.170, 0.150, and 0.150 for FOV=2.0, 1.5, 1.0, and 0.5, respectively. MATD falls to 0.010 in the final FOV=0.5 stage, showing that assignment pretraining by itself does not provide enough adaptation as visibility decreases. MATG is stronger than MATD and sometimes performs better on auxiliary indicators such as coverage or deterministic evaluation, but it does not match MATDP on the primary flow success-rate metric. A separate fixed-FOV comparison against MAPPO further checks the external baseline under the same PettingZoo environment reward, with the MATDP auxiliary reward terms disabled. In that aligned comparison, MATDP obtains higher last-100 reward in FOV=2.0, FOV=1.5, and FOV=0.5 and lower worst-case landmark distance in all four FOV settings, while MAPPO remains competitive at FOV=1.0 and on some sparse success indicators. The empirical claim of this thesis should therefore remain precise: MATDP improves the main flow success profile and shows stronger reward and distance trends in the aligned MAPPO comparison, but it does not dominate every metric under every evaluation mode.

The failure analysis is equally important. In FOV=0.5, successful stochastic episodes still occur, but deterministic evaluation remains weak. Additional diagnostics with memory bits, structured memory, relaxed PPO settings, and stronger exploration do not remove the failure. These results suggest a structural limitation of decentralized execution under severe occlusion. When an actor rarely observes landmarks or teammates, the centralized critic can estimate a better value target during training, but it cannot supply missing information at execution time. Rare successful trajectories are therefore difficult for PPO to consolidate into a stable local policy.

2. Contributions

The main contributions of this thesis are summarized as follows.

- It formulates the experimental task as FOV-limited cooperative control under a Dec-POMDP view, with masked actor observations and unmasked critic observations following CTDE.
- It designs a MATDP actor architecture that combines per-agent history encoding, team-level cross-agent representation, 15-dimensional joint diffusion action generation, and PPO-based online adaptation.
- It adapts PPO to the joint diffusion policy through stored denoising chains and step-average diffusion log-probability ratios.
- It uses soft-assignment PBRS and assignment auxiliary learning to stabilize training without replacing the reinforcement-learning objective.
- It evaluates MATDP, MATD, and MATG through a four-stage FOV flow and identifies the unresolved deterministic failure mode at $\text{FOV}=0.5$.
- It adds an aligned fixed-FOV MATDP-MAPPO baseline comparison with auxiliary reward terms disabled, clarifying that the external-baseline advantage is strongest for reward and worst-case distance rather than success rate.

3. Potential Application Scenarios and Practical Use

MATDP is most suitable for cooperative control problems where several agents must make continuous actions from incomplete local observations. The target scenario is not a fully observable centralized controller, but a decentralized execution setting in which each agent sees only a local region and must still coordinate with others. Examples include autonomous-driving simulation with occlusions, multi-robot coverage or inspection tasks, and cooperative unmanned vehicles searching or patrolling an area. In these settings, the Transformer history encoder can summarize recent local evidence, while the joint diffusion policy can model coordinated action patterns across agents.

A practical use of MATDP would follow a staged workflow. First, the target task should be formulated as a Dec-POMDP by defining the agents, continuous action space, local observation vector, team reward, and success condition. Second, the actor input should be converted into FOV-masked observation histories so that deployment-time information constraints are respected during training. Third, a centralized critic can be trained with richer or unmasked information under the CTDE principle, while the actor remains restricted to its local history. Fourth, training can start from an easier visibility setting and then gradually reduce the FOV radius through a flow. Finally, reward shaping and assignment auxiliary learning can be used as training-time stabilizers, but the deployed policy should use only the actor-side inputs available at execution time.

The framework should not be interpreted as a ready-to-deploy controller for real vehicles or robots. The present experiments are conducted in the compact MPE `simple_spread` environment, which isolates

coordination under partial observability but does not include sensor noise, vehicle dynamics, collision-risk constraints, road geometry, or real-time safety requirements. For practical deployment, MATDP would need to be combined with perception modules, safety filters, communication protocols, and domain-specific dynamics models. The current contribution is therefore best understood as a learning framework for studying FOV-limited coordination, rather than as a complete application system.

4. Limitations and Future Work

The results should be interpreted with several limitations in mind. The main flow is designed as a controlled comparison rather than an exhaustive benchmark study. This design is useful for isolating how Transformer history encoding, joint diffusion action generation, and PPO adaptation affect FOV-limited coordination under the same training budget. However, the evidence is still limited to one compact cooperative-control environment and one flow design. The reported improvement should therefore be understood as evidence for this experimental setting, not as a general claim about all partially observable multi-agent tasks. MPE `simple_spread` is useful for isolating coordination under partial observability, but it does not contain the perception noise, dynamics, safety constraints, heterogeneous agents, or road geometry that would appear in real autonomous-driving scenarios.

The aligned MAPPO comparison reduces one important fairness concern, because both algorithms are evaluated with the same PettingZoo environment reward and the auxiliary MATDP reward terms are disabled.

Nevertheless, it is still a fixed-FOV baseline on the same compact benchmark, not a broad benchmark suite. The MAPPO result also shows that the advantage is metric-dependent: MATDP has better reward and final worst-case distance trends, but the success rate remains sparse and mixed.

The most important technical limitation is the absence of explicit information exchange between actors. The experiments show that history encoding and joint diffusion action generation can improve stochastic success, but they are not enough when visible evidence becomes extremely sparse. Future work should therefore prioritize better information flow. Communication modules or message-passing policies could allow agents to share compact information about recently observed landmarks and teammate positions, following communication-based MARL directions such as DIAL [14]. Stronger learned memory, belief-state prediction, or contrastive state reconstruction may also help the actor retain useful evidence after objects leave the FOV.

The training pipeline can also be strengthened. Longer training budgets, broader flow schedules, and richer evaluation environments would be useful before making benchmark-level claims. Diffusion pretraining from expert, search-based, or scripted trajectories may help the policy consolidate rare successful behaviors before PPO fine-tuning. Finally, the framework should be tested in richer multi-agent driving or robotics simulators where occlusion, safety, and communication constraints are closer to practical deployment.

In conclusion, MATDP shows that Transformer-based history modeling, joint diffusion action generation, and PPO adaptation can work together for partially observable multi-agent control. The aligned

MAPPO comparison strengthens the reward-level evidence under a fairer external baseline, while the sparse success results keep the claim bounded. The same experiments also show where this combination breaks: when the actor's local observations contain too little task-relevant evidence, stronger optimization alone is not enough. Robust decentralized coordination under severe visibility constraints will likely require both better policies and better information sharing.

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Abstract

Transformer-Diffusion-PPO를 이용한 다중 에이전트 강화 학습 연구

다중 에이전트 강화 학습은 협력형 자율 시스템을 구현하기 위한 핵심 기술이지만, 각 에이전트가 제한된 국소 관측만을 이용하여 연속 제어 행동을 결정해야 하는 경우 안정적인 학습이 어렵다. 본 논문은 시야각(Field-of-View, FOV)이 제한된 다중 에이전트 협력 제어 문제를 대상으로, 관측 이력 모델링과 공동 행동 생성을 결합한 Multi-Agent Transformer-Diffusion-PPO (MATDP) 강화 학습 프레임워크를 제안한다.

제안한 방법은 문제를 분산 부분 관측 마르코프 결정 과정(Dec-POMDP)으로 정식화한다. 실행 단계의 actor는 FOV 마스크가 적용된 국소 관측 이력만을 입력으로 사용하고, 학습 단계의 critic은 중앙집중식 학습-분산 실행(CTDE) 원칙에 따라 마스크가 적용되지 않은 관측을 사용한다. Actor는 각 에이전트의 관측 이력을 Transformer encoder로 표현한 뒤, team-level Transformer를 통해 에이전트 간 문맥 정보를 통합한다. 이후 joint diffusion policy가 세 에이전트의 15차원 공동 행동을 한 번에 생성한다. 정책은 Proximal Policy Optimization (PPO)으로 온라인 업데이트되며, diffusion step별 log probability, 중앙집중식 value function, potential-based reward shaping, assignment auxiliary objective를 함께 사용하여 학습 안정성을 높인다.

실험은 세 개의 에이전트와 세 개의 landmark로 구성된 MPE `simple_spread` 환경에서 수행하였다. 주요 평가는 actor의 FOV 반경을 2.0에서 1.5, 1.0, 0.5로 점차 줄이는 네 단계 FOV 학습 설정으로 구성하였다. 동일한 학습 규모와

FOV 설정에서 제안한 MATDP, 온라인 PPO 업데이트를 제거한 MATD, 그리고 diffusion policy를 Gaussian policy로 대체한 MATG를 비교하였다. 주요 실험에서 MATDP는 모든 FOV 단계에서 final 100 episodes의 평균 success rate가 가장 높았으며, 각 단계의 성공률은 각각 0.190, 0.170, 0.150, 0.150이었다. MATD는 가장 어려운 FOV=0.5 단계에서 크게 성능이 저하되어, 온라인 정책 적응의 필요성을 보여준다. MATG와 비교했을 때도 MATDP는 통제된 네 단계 FOV 비교에서 가장 안정적인 success-rate profile을 보였다.

한편 FOV=0.5 조건은 결정론적 관점에서 완전히 해결되지 않았다. MATDP는 확률적 탐색 과정에서는 성공 episode를 생성할 수 있었지만, deterministic evaluation에서는 여전히 낮은 성능을 보였다. 추가 진단 실험에서도 actor-side memory와 더 강한 exploration은 제한적인 개선만을 제공하였다. 따라서 본 연구는 Transformer-Diffusion 기반 정책이 부분 관측 다중 에이전트 제어에 유망함을 보이는 동시에, 극단적인 관측 제한 상황에서는 명시적 통신, 더 강한 memory mechanism, 그리고 더 풍부한 평가 환경이 필요함을 확인하였다.

주요어: 다중 에이전트 강화 학습, Transformer, diffusion policy, Proximal Policy Optimization, 부분 관측, 중앙집중식 학습-분산 실행